TASK 3: INHERITANCE

Python OOP Practical Report — 20 Marks

1. Objective

To create a parent class and a child class, reuse common attributes and methods, and add child-specific functionality.

2. OOP Concept Used

Inheritance, parent class, child class, super(), and method reuse.

3. Source Code

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)

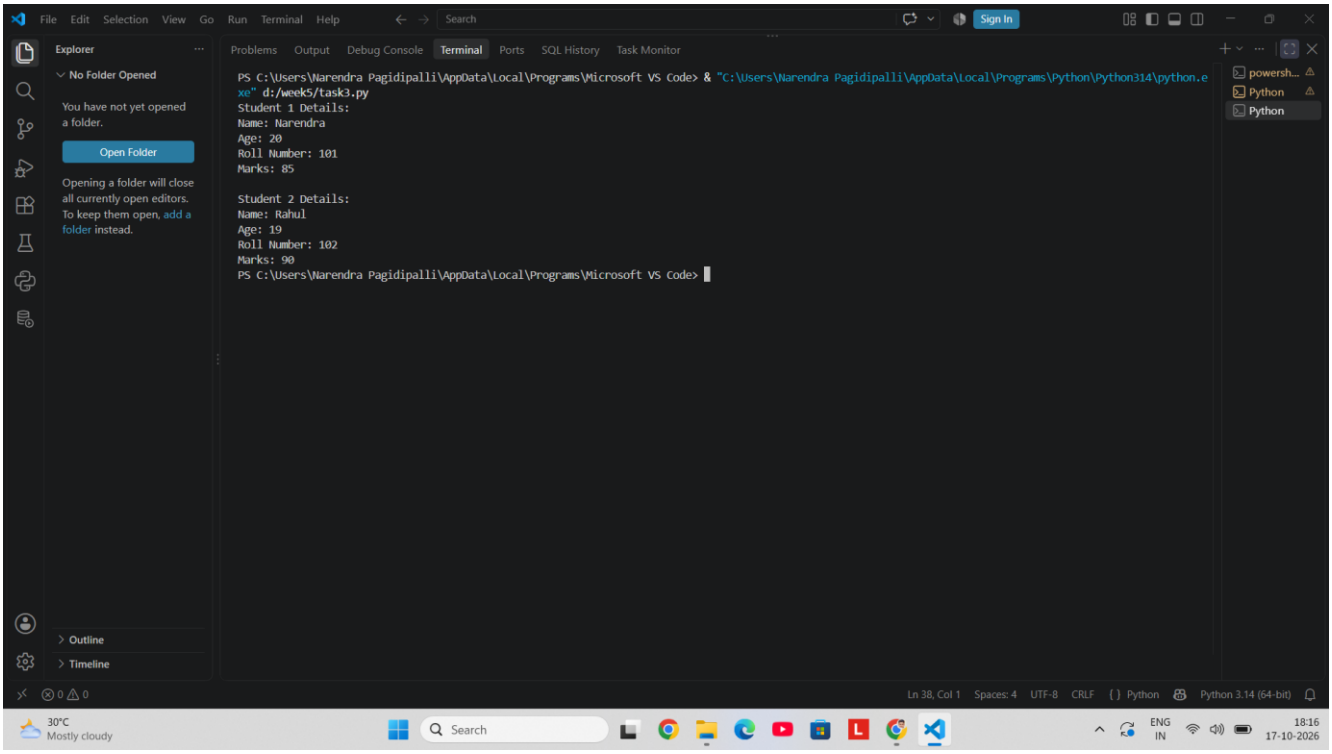

class Student(Person):
    def __init__(self, name, age, roll_no, marks):
        super().__init__(name, age)
        self.roll_no = roll_no
        self.marks = marks

    def display_student(self):
        self.display()
        print("Roll Number:", self.roll_no)
        print("Marks:", self.marks)


s1 = Student("Narendra", 20, 101, 85)
s2 = Student("Rahul", 19, 102, 90)

print("Student 1 Details:")
s1.display_student()
print("\nStudent 2 Details:")
s2.display_student()
```

4. Output



5. Explanation

- Person is the parent class and contains the common name and age attributes.
- Student inherits the attributes and display() method from Person.
- super().__init__(name, age) calls the parent constructor to initialize common attributes.
- Student adds its own roll_no and marks attributes.
- The objects s1 and s2 display both inherited and child-specific information.

6. Learning Outcome

I learned how a child class can inherit functionality from a parent class and extend it with additional attributes and methods.

7. Conclusion

This program demonstrates inheritance and code reuse using the Person and Student classes.